

Distributed Systems Exam, Feb 2, 2021

Exam duration: 60 minutes

Questions:

- 30 questions from Distributed Systems course x 3 points each;
- 1 required question of 10 points if you answer "Yes" to the respective question;
- 1 required question regarding entering your ID (group and name);

Question types:

- Radio-button: only one answer is correct;
- Checkbox: one or more (including all) answers are correct; no partial credit;

Exam mark: Total earned points / 10;

...

Punkte: 100/100

1. Which of the followings are true about vertical fragmentation of a table in distributed storage design?

(3/3 Punkte)

- ☐ Only one of the fragments must include the primary key column(s) of the original table
- ☒ All fragments must include the primary key column(s) of the original table. ✓
- ☐ The fragments may not include the primary key column(s) of the original table.
- ☒ Vertical fragmentation must allow table re-construction from the fragments ✓

2. For which type of logical clocks the event timestamp must include the processor index for having a totally ordered events in a distributed system?
(3/3 puncte)

- ☒ Scalar time ✓
- ☐ Vector time
- ☐ Both scalar time and vector time

3. Which of the followings are true about atomic commit protocols in distributed transactions?
(3/3 puncte)

- ☒ Each Participant must call the join() operation defined by the Coordinator interface; ✓
- ☐ The Coordinator must call the join() operation defined by Participants' interface;
- ☒ In a distributed transaction, the Coordinator must have the references to all Participants; ✓

4. Check all true statements about Virtual Machines (VMs).
(3/3 puncte)

- ☐ All VMs running on the same server must have the same guest Operating System;
- ☒ VMs can be replicated in multiple servers; ✓
- ☒ VM instances may be created from a template VM; ✓
- ☐ The virtualization layer provides services for data exchange between VMs;

5. Which of the followings are specific only to distributed deadlocks?
(3/3 puncte)

- ☐ Deadlock situations are detected by using wait-for graphs;
- ☐ Deadlock situations are detected by using timeout timers;

☒ A phantom deadlock situation may happen during a deadlock detection process; ✓

6. Concurrency control is the main task of:
(3/3 puncte)

- ☐ Recovery Manager
- ☐ Data Manager
- ☒ Scheduler ✓
- ☐ Transaction Manager

7. Which of the following is the correct definition of the reliable one to one communication:
(3/3 puncte)

- ☐ Any message in the outgoing message buffer is delivered to the incoming message buffer
- ☐ The message received is identical to one sent, and no messages are delivered twice
- ☒ Any message in the outgoing message buffer is delivered to the incoming message buffer and the message received is identical to one sent, and no messages are delivered twice ✓

8. The transparency type that offers the same interface to a service, no matter if it is local or remote is called:
(3/3 puncte)

- ☒ Access transparency ✓
- ☐ Location transparency
- ☐ Migration transparency
- ☐ Concurrency transparency

9. DEFAULT POINTS QUESTION. Do you want the default 10 points? *
(10/10 puncte)

☒ YES ✓

10. Check all true statements about a Global Conceptual Schema (GCS)
(3/3 puncte)

☐ Includes how distributed tables are fragmented

☐ Includes storage locations for distributed tables

☐ Includes information about tables replication

☒ Includes all primary keys of the tables ✓

☒ Includes all foreign keys of the tables ✓

11. Google App Engine (GAE) is a type of:
(3/3 puncte)

☐ Infrastructure as a Service

☐ Software as a Service

☒ Platform as a Service ✓

☐ None of the indicated answers

12. (True or False) Failure is easier to be detected for synchronous distributed systems than for asynchronous distributed systems
(3/3 puncte)

☒ True ✓

☐ False

13. In which type of virtualization technique the guest operating system needs to be modified to be adapted to the hypervisor?
(3/3 puncte)

- ☐ Full Virtualization of type 1
- ☐ Full Virtualization of type 2
- ☒ Para-Virtualization ✓
- ☐ Hardware-assisted Virtualization;

14. (True or False) In a distributed system, strong consistency of replicated data has a positive impact over latency.
(3/3 puncte)

- ☐ True
- ☒ False ✓

15. Check all true statements about a RPC client stub
(3/3 puncte)

- ☒ It should be compiled and linked with the client code ✓
- ☐ It should be compiled and linked with the server code
- ☐ It should implement the remote procedure
- ☒ It is used to achieve the goal of semantic transparency ✓

16. Check all true statements about a binding agent used in client - server binding process;
(3/3 puncte)

- ☒ Binding agent location should be known by both servers and clients; ✓

- ☐ Binding agent location should be known only by the clients;
- ☐ Communication overhead is low when most of the clients are short lived;
- ☒ Replicated binding agents involves high overhead; ✓
- ☐ Replicated binding agents involves low overhead;

17. An interprocess communication where both send and receive operations are blocking operations is called:
(3/3 puncte)

- ☐ Transient communication
- ☐ Persistent communication
- ☒ Synchronous communication ✓
- ☐ Asynchronous communication

18. Which of the following statements are true about Chandy-Lamport snapshot algorithm?
(3/3 puncte)

- ☐ It works for all types of communication networks;
- ☒ It works only for specific communication networks; ✓
- ☐ The marker must be sent by a node before taking its snapshot
- ☒ The marker must be sent by a node after taking its snapshot ✓
- ☐ A node may send the marker before or after taking its snapshot;

19. Which of the followings are true about table fragmentation in distributed storage design:
(3/3 puncte)

- ☐ table fragments must not be replicated
- ☐ table fragments must be stored at different sites
- ☒ table fragmentation increases data processing parallelism ✓
- ☒ table fragmentation helps in case of disaster recovery ✓

20. Which of the followings are true about nested distributed transactions?
(3/3 puncte)

- ☒ Children sub-transactions may run on different machines; ✓
- ☐ Children sub-transactions must run on the same machine;
- ☒ Children sub-transactions may run in parallel; ✓
- ☐ For consistency reasons, children sub-transactions must be executed one after the other;

21. Consider a transaction T operating on an object x and transaction U operating on an object y.
Check all true statements below.
(3/3 puncte)

- ☐ T read / U read is a conflicting situation;
- ☐ T read / U write is a conflicting situation;
- ☐ T write/ U read is a conflicting situation;
- ☐ T write / U write is a conflicting situation;
- ☒ The mentioned operations are not conflicting situations; ✓

22. Check the true statement.
(3/3 puncte)

- ☐ Asynchronous distributed system failures can be detected by using timeout timers;
- ☒ Synchronous distributed system failures can be detected by using timeout timers; ✓
- ☐ Timeout timers can be used for failure detection in both synchronous and asynchronous distributed systems

23. Which of the followings is true about a flat distributed transaction?
(3/3 puncte)

- ☒ Each transaction operation must be completed before executing another operation; ✓
- ☐ Each transaction operation must be executed at a different site;
- ☐ Transaction operations may be executed in parallel;

24. The unwanted situation that may happen, when the acquired locks are progressively released before transaction execution completion is called:
(3/3 puncte)

- ☐ Lost update situation
- ☐ Inconsistent retrieval situation
- ☒ Cascading abort situation ✓
- ☐ Deadlock situation

25. Enter your ID according to the following syntax: <Group> <Name> <FirstName>, <Year of taking the cours>, <Teaching Assistant Name> *

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26. Which of the followings is true about the events in a distributed system?
(3/3 puncte)

- ☐ The timestamp of an event represents the physical time when the event happened;
- ☐ Concurrent logical events must occur at the same physical time
- ☒ Logically concurrent events are not causally related events; ✓

27. Which ACID property describes the fact that all transaction effects should be saved into permanent storage after the successful transaction commit.
(3/3 puncte)

- ☐ Consistency
- ☐ Isolation
- ☒ Durability ✓
- ☐ Atomicity

28. Which of the followings are features of Optimistic Concurrency Control (OCC) technique?
(3/3 puncte)

- ☒ It is recommended for private workspace transaction implementations; ✓
- ☐ It is recommended for write-ahead log transaction implementations;
- ☐ There are no conflicts in OCC, hence all concurrent transactions may commit without any validation;
- ☒ In OCC all transactions must be validated before they are allowed to commit. ✓

29. Which of the followings represent benefits of resource virtualization?
(3/3 puncte)

- ☒ Server consolidation ✓
- ☒ Elastic usage of hardware resource ✓

- ☐ Avoids single points of failure
- ☐ Limited infrastructure and resource demand

30. (True or False) Data replication improves data consistency.
(3/3 puncte)

- ☐ True
- ☒ False ✓

31. The ability of a distributed system to limit the latency due to system faults is called:
(3/3 puncte)

- ☐ Reliability
- ☐ Consistency
- ☒ Availability ✓
- ☐ Efficiency

32. Which of the following statements are true about the atomic Two-Phase Commit Protocol (2PC) for Distributed Transactions?
(3/3 puncte)

- ☒ The participants must save the involved transaction data into permanent storage before voting commit; ✓
- ☐ The participants must save the involved transaction data into permanent storage after voting commit;
- ☒ Each protocol step defines a timeout timer for avoiding algorithm blocking; ✓
- ☐ The coordinator may decide to commit the transaction even if a participant voted abort;

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